I'll start by showing you how to create a basic ASP.NET Core application using one of the Visual Studio templates. If you're using other tooling, such as the .NET command-line interface (CLI), you'll have similar templates available. I use Visual Studio 2022 and ASP.NET Core 7 with .NET 7 in this chapter, but I also provide tips for working with the .NET CLI.

TIP You can view the application code for this chapter in the GitHub repository for the book at http://mng.bz/5wj1.

After you've created your application, I'll show you how to restore all the necessary dependencies, compile your application, and run it to see the output. The application will be simple, containing the bare bones of an ASP.NET Core application that responds with `"Hello World!"`

Having run your application, your next step is understanding what's going on! We'll take a journey through the ASP.NET Core application, looking at each file in the template in turn. You'll get a feel for how an ASP.NET Core application is laid out and see what the C# code for the smallest possible app looks like.

As a final twist, you'll see how to extend your application to handle requests for static files, as well as how to create a simple API that returns data in standard JavaScript Object Notation (JSON) format.

At this stage, don't worry if you find parts of the project confusing or complicated; you'll be exploring each section in detail as you move through the book. By the end of the chapter, you should have a basic understanding of how ASP.NET Core applications are put together, from when your application is first run to when a response is generated. Before we begin, though, we'll review how ASP.NET Core applications handle requests.

3.1 A brief overview of an ASP.NET Core application